Dachuan Liu

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Research interests: Multi-Agent Path Finding (MAPF), Multi-Robot Planning & Coordination, Self-Reconfigurable Modular and Swarm Robotics

Education

University of Michigan Ann Arbor, MI
B.S.E. in Computer Engineering | GPA: 3.91/4.00 *Expected May 2027*
- Relevant coursework: Machine Learning, Robot Kinematics and Dynamics, Self-Driving Cars: Perception and Control

Shanghai Jiao Tong University, UM-SJTU Joint Institute Shanghai, China
B.S.E. in Mechanical Engineering (Dual Degree) | GPA: 3.73/4.00 *Expected Aug 2027*
- Relevant coursework: Design and Manufacturing, Mechanical Laboratory, Differential Equations and Linear Algebra

Publications

[C1] **D. Liu**, S. Zhou, S. Zhao, Z. Ren. "LSRP*: Scalable and Anytime Planning for Multi-Agent Path Finding with Asynchronous Actions." *Under review*, IEEE International Conference on Robotics and Automation (ICRA), 2027. arXiv:2611.10363.

Academic Service

- Sub-reviewer, *IEEE Transactions on Intelligent Transportation Systems* (T-ITS), 2026

Research Experience

Robotics, Autonomy, and Planning Lab, Shanghai Jiao Tong University Shanghai, China
Research Assistant | Advisor: Prof. Zhongqiang Ren *Apr 2026 – Present*
- Developed **LSRP***, an anytime C++ solver for MAPF with Asynchronous Actions (MAPF-AA) that uses dominance pruning and restart strategies to keep improving solution quality within the time budget. Scales to 1,000 agents and reduces sum of costs by over 40% compared with LSRP; first-author paper submitted to ICRA 2027.
- Developed **LSRP*-LA**, extending LSRP* to large agents with 1×1 to 3×3 footprints, with a swept-area occupancy model and a rule layer that uses recursive push trees and lane evacuation to improve scalability and solution quality. Scales beyond 800 agents and reduces sum of costs by up to 62% compared with Layered LA-LaCAM.

Self-Reconfigurable Modular Robotics, Robotics Department, University of Michigan Ann Arbor, MI
Undergraduate Researcher (ROB 490) | Advisor: Prof. Steven Ceron *Sep 2026 – Present*
- Finalizing the design of, and fabricating, a fleet of 20–25 cylindrical modular robots that attach to neighbors with an extendable, rotating internal magnet and reconfigure by rolling around each other's perimeters; working with PhD student Tianyi Hu.
- Diagnosed a stall in the self-rotation drive (insufficient motor torque under shell friction) and specified a dual high-torque motor configuration.
- Redesigned the onboard power system after identifying that a 3.7 V cell could not meet the motor's 1.47 A stall current; selected a 7.4 V 2S LiPo with a 3 A DC-DC buck converter regulated to 5–6 V.
- Ran 200-cycle fatigue tests of the magnet extension/retraction gear mechanism (no observable wear) and fabricated a replacement lead screw; module assembly time is 2–3 hours each.

Multidisciplinary Design Program, University of Michigan Ann Arbor, MI
Simulation Engineer | Advisor: Prof. Paul Green *Jan 2026 – Present*
- Designed driving-simulation experiments in Unreal Engine 5 that vary obstacle type and curve radius, and analyze drivers' visual demand using the visual occlusion method.
- Running participant experiments and data analysis for a manuscript in preparation.

Software Engineering Research Group, Shanghai Jiao Tong University Shanghai, China
Software Engineer | Advisor: Prof. Yan Liu *Feb 2024 – Oct 2024*
- Built a full-stack interactive teaching platform (HTML/CSS/JS, MySQL) for teacher–student interaction and research data management; implemented the student-side client, which received a registered software copyright.

Teaching Experience

EECS 216: Introduction to Signals and Systems, University of Michigan Ann Arbor, MI
Instructional Aide *Sep 2026 – Dec 2026*
- Lead 5 lab sections (30 students each), hold weekly office hours, and grade lab reports and exams.

ECE 216: Introduction to Signals and Systems, UM-SJTU Joint Institute Shanghai, China
Teaching Assistant | Instructor: Prof. Heng Qiao *May 2026 – Aug 2026*

- Taught twice-weekly recitations and weekly office hours; led 3 lab sessions (30 students) and graded assignments, lab reports, and exams.

MATH 286: Differential Equations and Linear Algebra, UM-SJTU Joint Institute Shanghai, China
Teaching Assistant | Instructor: Prof. Horst Hohberger *Sep 2025 – Dec 2025*

- Taught weekly recitations covering linear algebra, ODEs, Fourier series/transforms, and Laplace transforms; graded weekly assignments, four quizzes, and two exams.

Selected Projects

Learning Priority Orders for Prioritized SIPP in MAPF-AA (ECE 7606) Shanghai, China
C++, Reinforcement Learning *May 2026 - Aug 2026*

- Trained a Plackett-Luce ranking policy with REINFORCE over per-agent conflict features to order agents for prioritized SIPP in MAPF with asynchronous actions; on held-out four maps (den312d, Room64, Warehouse, Random32) instances, raised success rate at 60 agents from 35% to 65% and the largest solved team from 80 to 140 agents.

Drone Delivery Route Optimization (EECS 281) University of Michigan
C++ *Sep 2025 – Dec 2025*

- Solved TSP-style delivery routing on graphs: Prim's MST for connectivity, branch-and-bound with pruning for exact solutions, and arbitrary insertion + 2-opt for near-optimal solutions under strict runtime limits.

Column-Climbing Robot (ME 350) UM-SJTU Joint Institute
SolidWorks, Raspberry Pi, Python *May 2025 – Aug 2025*

- Designed and fabricated a robot that climbs columns of varying diameter; invented a bicycle-clamp locking mechanism that holds position without power, and built the motor and laser-sensor control loop.

Skills

Programming: C++, Python, C, MATLAB, SQL, HTML/CSS/JavaScript
Robotics & Tools: Simulink, PyTorch, Unreal Engine 5, Git, LabVIEW, Mathematica, Flask
Hardware: SolidWorks, Arduino, Raspberry Pi, NI myDAQ
English: TOEFL iBT 5.5/6

Honors and Awards

JI Student Development Scholarship, UM-SJTU Joint Institute 2023 – 2024
Outstanding Student Scholarship (Class B), Shanghai Jiao Tong University 2023 – 2024
Registered Software Copyright (teaching platform, student client) Sep 2024
Basic Teaching Assistant Certificate, UM-SJTU Joint Institute Aug 2024